RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Thursday, 1 October 2026, 10:50 AM
State	Finished
Completed on	Thursday, 1 October 2026, 11:11 AM
Time taken	21 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int main(){
    int t,n1,n2,i,j,first;
    scanf("%d",&t);
    while(t--){
        scanf("%d",&n1);
        int *a=malloc(n1*sizeof(int));
        for(i=0;i<n1;i++){
            scanf("%d",&a[i]);
        }
        scanf("%d",&n2);
        int *b=malloc(n2*sizeof(int));
        for(i=0;i<n2;i++){
            scanf("%d",&b[i]);
        }
        i=0;
        j=0;
        first=1;
        while(i<n1 && j<n2)
        {
            if(a[i]==b[j]){
                    if(first==0)
                        printf(" ");

                    printf("%d",a[i]);
                    first=0;

                    while(i<n1 && a[i] == a[i+1] && i+1<n1){
```

```
                    i++;
                }
                while(j<n2 && b[j]==b[j+1] && j+1<n2){
                    j++;
                }
                i++;
                j++;
            }
            else if(a[i]<b[j])
            {
                i++;
            }
            else{
                j++;
            }
        }
        printf("\n");

        free(a);
        free(b);
    }
    return 0;

```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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